Lakmali Nadeesha Kumari

Machine Learning Research Scientist – Computer Vision, Generative AI, Foundation Models – Davis, CA
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RESEARCH SUMMARY

Machine learning researcher (Ph.D. 2026, now a postdoc at UC Davis) developing **data-efficient and robust deep learning** for large-scale vision problems. Research spans **foundation-model adaptation**, **generative models** (diffusion, variational, normalizing flows), **active and self-supervised learning**, and **agentic vision–language systems**, with first-author papers at MICCAI, IEEE ICIP, and Elsevier JMS. Brings industry research experience from **two Applied Scientist internships at Amazon** (generative AI and multimodal LLM pipelines) and 3+ years of engineering at Synopsys.

EDUCATION

Ph.D., Electrical and Computer Engineering (GPA 3.8/4.0) 2021 – 2026
University of Kentucky Advisor: Prof. Sen-ching Samson Cheung *Lexington, KY*
Dissertation: *Resource-Efficient Machine Learning Through the Balance of Prior Knowledge and Learned Structure*

B.Sc. Eng., Electrical and Electronic Engineering 2013 – 2017
University of Peradeniya *Sri Lanka*

RESEARCH AND INDUSTRY EXPERIENCE

Postdoctoral Researcher, AI for Digital Pathology Oct. 2026 – Present
University of California, Davis with Prof. Chen-Nee Chuah and Prof. Brittany N. Dugger *Davis, CA*

- Developing **agentic vision–language** and foundation-model systems for scalable, quantitative analysis of gigapixel whole-slide brain images for Alzheimer's disease research.
- Building whole-slide image pipelines that generalize across institutions, scanners, and stains on multi-GPU HPC.

Applied Scientist Intern (return internship) May 2025 – Aug. 2025
Amazon, Merch on Demand *Seattle, WA*

- Designed an end-to-end **generative AI pipeline** for automated, context-aware product visualization using diffusion models (FLUX.1, CatVTON-Flux) with **LoRA**-based refinement.
- Introduced an **LLM-as-a-judge** evaluation stage that automatically scores and filters generated images, closing the refinement loop without manual review; trained on **multi-GPU AWS SageMaker**.

Applied Scientist Intern Jun. 2024 – Aug. 2024

- Built a **multi-modal, multi-label classifier** on CLIP image–text embeddings for large-scale product data, significantly outperforming keyword-based baselines.
- Automated labeling with an **active-learning loop** and **LLM-based pseudo-labeling**, reducing manual annotation.

Graduate Research Assistant, Multimedia Information Analysis Lab Aug. 2022 – Sep. 2026
University of Kentucky Advisor: Prof. Sen-ching Samson Cheung *Lexington, KY*

- **Learning class difficulty [C2]:** proposed Dynamic Focal Attention, a learned per-class attention bias for SAM-based segmentation that improves difficult classes by **up to 15.2 Dice points** on gigapixel histopathology (BDSA, BCSS, CRAG) while removing a separate pre-training stage.
- **Label-efficient learning [C1]:** task-aligned active learning that re-derives its acquisition space from the evolving encoder, reaching **98.7%** of fully supervised Dice with **38.4%** of the labels and solving the cold-start problem.
- **Latent structure discovery [U2]:** CONTINUUM, an amortized variational model of continuous disease trajectories that recovered five reproducible subtypes across two independent cohorts, confirmed by replicated genetic associations.
- **Diffusion models [U1]:** temporally adaptive spectral control for removing unwanted concepts from diffusion models (targeting CVPR 2027).
- **Foundation models and video [A1]:** adapted SAM-Path to multicompartment brain-slide segmentation (vs. Path-SAM2, SegFormer); built a gaze-tracking video pipeline with the UC Davis MIND Institute.

Graduate Research Assistant, Augmented Intelligence for Smart Manufacturing Lab Aug. 2021 – Aug. 2022
University of Kentucky Advisor: Prof. Peng Wang *Lexington, KY*

- **Amortized Bayesian inference [J3]:** conditional invertible neural network trained on simulated data that replaces MCMC, **over 3,000× faster** (690 s → 0.2 s) with better prognosis accuracy.

Applications Engineer II Apr. 2018 – Aug. 2021
Synopsys *Sri Lanka*

- Component owner for SpyGlass Analyzer; validated static, CDC/RDC, and design-constraint verification EDA tools used by semiconductor customers. Received Synopsys recognition awards.

PUBLICATIONS

Peer-Reviewed Conference Papers

[C2] **R.M.L.N. Kumari**, S.-C. S. Cheung. "Learning Class Difficulty via Dynamic Focal Attention for Histopathology Segmentation." *MICCAI*, LNCS vol. 16891, Springer, 2026. [PDF]

[C1] **R.M.L.N. Kumari**, C. T. Bandara, C.-N. Chuah, S.-C. S. Cheung. "A Warmer Start to Active Learning with Adaptive Gaussian Mixture Models for Skin Lesion Segmentation." *IEEE ICIP*, pp. 2247–2252, 2025. [DOI]

Journal Articles

[J3] **R.M.L.N. Kumari**, P. Wang. "Efficient Stochastic Parametric Estimation for Lithium-Ion Battery Performance Degradation Tracking and Prognosis." *Journal of Manufacturing Systems*, vol. 75, pp. 270–277, 2024. [DOI]

[J2] **R.M.L.N. Kumari**, C. T. Bandara, M. B. Dissanayake. "Sylvester Matrix Based Similarity Estimation Method for Automation of Defect Detection in Textile Fabrics." *Journal of Sensors*, 2021. [DOI]

[J1] C. T. Bandara, **R.M.L.N. Kumari**, D. H. S. Maithripala, D. A. A. C. Ratnaweera. "Vehicle-Fixed-Frame Adaptive Controller and Intrinsic Nonlinear PID Controller for Attitude Stabilization of a Complex-Shaped Underwater Vehicle." *Journal of Mechatronics and Robotics*, vol. 4, 2020. [DOI]

In Preparation

[U2] **R.M.L.N. Kumari**, J. Zhu, R. K. Shahidehpour, P. T. Nelson, S.-C. S. Cheung. "CONTINUUM: A Variational Deep-Generative Reformulation of Subtype and Stage Inference Reveals Reproducible, Genetically Distinct Neuropathological Trajectories of Alzheimer's Disease Across Independent Autopsy Cohorts." For *Neurobiology of Disease*.

[U1] **R.M.L.N. Kumari**, C. T. Bandara, S.-C. S. Cheung. "Not All Steps Matter: Temporally Adaptive Spectral Control for Concept Removal in Diffusion Models." For *IEEE/CVF CVPR 2027*.

Published Abstracts

[A1] T. Pearce, L. Nadeesha, A. Neltner, *et al.* "Multicompartment Neuroanatomic Segmentation of Autopsy Brain Tissue Sections Using Deep Learning Methodologies." *Journal of Neuropathology & Experimental Neurology*, 84(6):494, 2025.

TECHNICAL SKILLS

Languages	Python, C/C++, SQL, MATLAB, R, Bash
Frameworks	PyTorch, TensorFlow, Hugging Face, PEFT/LoRA, CLIP, Segment Anything (SAM), FLUX.1
Research Areas	Computer vision, segmentation, diffusion models, LLMs and VLMs, multimodal learning, agentic AI, active and self-supervised learning, Bayesian and variational inference, normalizing flows
Infrastructure	AWS (SageMaker, Bedrock, EC2), multi-GPU distributed training, HPC, Docker, Git, Linux

SELECTED PRESENTATIONS

- "SemiSAM_Med: Class-Imbalanced Active Learning in Medical Image Segmentation via a Semi-Supervised SAM Model." ECE Research Symposium, University of Kentucky, 2024. [Poster]
- "Eyes Speak Louder: Computer Vision and Machine Learning Insights into Autism Spectrum Behaviors." Commonwealth Computational Summit, 2023. [Poster]
- "SAL-SMIS: Self-Supervised Active Learning for Semi-Supervised Medical Image Segmentation." ECE Research Symposium, University of Kentucky, 2023. [Poster]
- "Modeling and Prediction of Lithium-Ion Battery Degradation with CINN for Parameter Estimation." Engineering Dean's Advisory Council, University of Kentucky, 2023. [Poster]

LEADERSHIP, TEACHING, AND HONORS

Assistant Director, Graduate Society of Women Engineers (GradSWE), University of Kentucky	2023 – 2024
Graduate Teaching Assistant, Electrical and Computer Engineering, University of Kentucky	2022 – 2023
Vice President (Membership), Synopsys Lanka Toastmasters Club	2020
Synopsys recognition awards for technical contributions	2019 – 2021
Top 1% of ∼250,000 candidates nationally, GCE Advanced Level, Sri Lanka	2012